

RECRUITAI

PROJECT SPECIFICATION & IMPLEMENTATION BLUEPRINT

IoT Sensor Dashboard (React)

Detailed project brief generated from the project record.

CANDIDATE

Rohan Desai

DURATION

3 days

TECHNOLOGY STACK

Java

HTML

CSS

React

Project Overview

Summary of the problem, purpose and expected outcome.

PROJECT SUMMARY

Create a simple React web page that visualises static IoT sensor data in cards. The app must be responsive and demonstrate basic React, HTML and CSS skills.

CANDIDATE Rohan Desai	TYPE frontend	DURATION 3 days
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Primary Objectives

- Implement a responsive layout using HTML and CSS.
- Build reusable React components for sensor cards.
- Load and render static JSON data via the Fetch API.
- Demonstrate basic version control with Git.

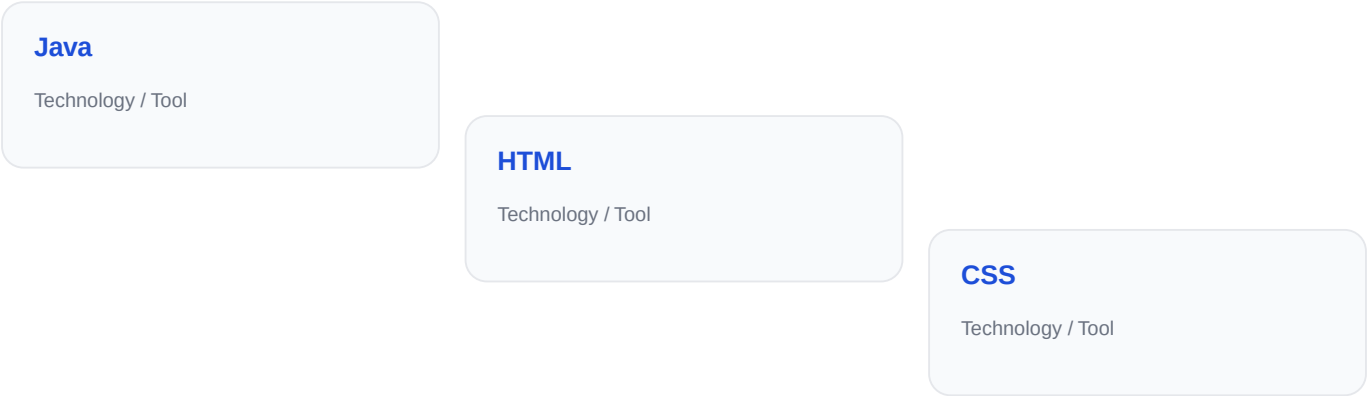
Candidate Details

NAME

Rohan Desai

Technology Stack

Technologies, frameworks and tools expected for implementation.



React

Technology / Tool

Functional Requirements

- Show a header with the dashboard title.
- Render a card for each sensor entry in sensors.json.
- Display sensor name, numeric reading and a colour coded status indicator.
- Provide a refresh button that reloads the JSON file.
- Show a loading spinner while data is being fetched.
- Handle fetch errors with a user friendly message.

Non-Functional Requirements

- Responsive design for screens >320 px.
- CSS must use BEM naming conventions.
- React code must use functional components and hooks.
- All user visible text must be in English.
- The app must load within 2 seconds on a typical broadband connection.

Modules & Features

Major functional areas that make up the project.

Layout

Overall page structure and responsive grid.

- Header
- Footer
- Responsive grid

Sensor Card

Reusable component showing sensor details.

- Name display
- Value display
- Status colour

Data Layer

Fetch JSON and provide data to components.

- Fetch wrapper
- Loading state
- Error handling

Implementation Roadmap

Recommended phases and milestones for completing the project.

1

Setup

Initialize Git repo., Run create react app., Add basic folder structure., Commit initial code.

- Initialize Git repo.
- Run create react app.
- Add basic folder structure.
- Commit initial code.

2

Core UI

Create Header and Footer components., Implement responsive grid in CSS., Develop SensorCard component with props., Add placeholder data to render three cards.

- Create Header and Footer components.
- Implement responsive grid in CSS.
- Develop SensorCard component with props.
- Add placeholder data to render three cards.

3

Data Integration

Add sensors.json in public folder., Write fetchData hook using useEffect., Show loading spinner while fetching., Display error message on failure., Wire fetched data to SensorCard components.

- Add sensors.json in public folder.
- Write fetchData hook using useEffect.
- Show loading spinner while fetching.
- Display error message on failure.
- Wire fetched data to SensorCard components.

4

Polish & Delivery

Refine CSS (BEM, mobile breakpoints)., Add refresh button to re fetch data., Run basic manual tests on different browsers., Update README with run instructions., Push final commit and tag release.

- Refine CSS (BEM, mobile breakpoints).
- Add refresh button to re fetch data.
- Run basic manual tests on different browsers.
- Update README with run instructions.
- Push final commit and tag release.

Evaluation Criteria

Criteria that can be used to assess the completed project.

- Criterion: Correct UI rendering | Description: All cards and layout appear as specified. | Weight: 20
- Criterion: Responsive design | Description: Layout adapts to mobile and desktop. | Weight: 20
- Criterion: Data handling | Description: Fetches JSON, shows loading and error states. | Weight: 20

- Criterion: Code quality | Description: Uses functional components, hooks, BEM CSS. | Weight: 15
- Criterion: Version control | Description: Git history shows logical commits. | Weight: 15
- Criterion: Documentation | Description: README explains setup and usage. | Weight: 10

Expected Deliverables

Artifacts and outcomes expected at project completion.

- Git repository URL
- Source code (HTML, CSS, JS/React)
- sensors.json file
- README with setup and run steps
- Demo screenshots
- Brief implementation notes

Candidate Alignment

Project requirements compared with the candidate information available in the project record.

Rohan already knows C++ and IoT concepts, which help him understand sensor data. This three day React project introduces frontend basics (HTML, CSS, React) while staying within his junior level and leveraging his existing knowledge of IoT data structures.

Implementation Note

This document is generated from the project record. AI-generated requirements should be reviewed by an administrator before implementation.